



Novel and old news sentiment in commodity futures markets

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ABSTRACT

This study investigates the relationship between novel and old news sentiment and commodity futures returns. Using TRNA data from Thomson Reuters, we measure daily sentiment of both novel and old news to estimate their impact on commodity futures returns. Our findings reveal that both novel and old news sentiment significantly correlate with returns, with old sentiment having a stronger effect. Notably, only old news sentiment triggers an overreaction on the news day, which largely reverses over the subsequent 30 trading days. During periods of high financial stress and uncertainty, old news sentiment has a more pronounced impact on commodity futures returns. This paper contributes to the literature by highlighting the distinct impact patterns of old and novel news sentiment.

1. Introduction

Literature has highlighted the effect of news on the dynamics of the financial market (Barber and Odean, 2008; Chan, 2003; García, 2013; Niederhoffer, 1971; Piñeiro-Chousa et al., 2016; Tetlock, 2007; Tetlock et al., 2008; Tetlock, 2011). News in the financial market is important for investors as it offers them timely updates on asset-related information. Consequently, investors integrate the information to adjust asset prices. However, asset prices do not always incorporate such information accurately or rationally. On the one hand, investors may underreact to new information, leading to observed drifts or delayed reactions of returns to the information (Fedyk, 2023; Hirshleifer and Teoh, 2003). On the other hand, investors may overreact to the information contained in news items, resulting in subsequent return reversals (García, 2013; Tetlock, 2007).

The investor inattention theory (Gilbert et al., 2012) or the correlation neglect theory (Enke and Zimmermann, 2017) suggests that investors may confuse old information with new information when these pieces are combined or when old information is combined with new information in a recently released news item. The existing literature primarily examines the effects of novel and old news on asset prices based on the volume of old information in news items (Fedyk and Hodson, 2023; Tetlock, 2011), with a limited focus on the sentiment of these two types of news. News sentiment has been long emphasized as one factor that can significantly explain the return movement of asset returns (García, 2013; Hillert et al., 2014; Loughran and McDonald,

2011; Tetlock, 2011, 2007). Therefore, this study focuses on the impact of novel and old news sentiment. We define old news items as those that contain information already released in previous news. Meanwhile, novel news refers to those that only contain new information.

In this study, our main focus is to examine the impact of both novel and old news sentiment on commodity futures returns. Commodity futures are particularly suitable for studying the influence of news sentiment, as previous research has shown that commodity futures returns are significantly affected by commodity-specific news sentiment (Abdollahi, 2023; Akyildirim et al., 2022; Li et al., 2021; Rogmann et al., 2024; Smales, 2015, 2014). Since information about commodities is sourced from various widely scattered locations around the world, media news plays an important role in providing timely updates to commodity investors.

This study uses Thomson Reuters News Analytics data (TRNA) to measure news sentiment. TRNA covers a wide range of common finance-related newswires, which are fed into a system called machine readable news (MRN). The MRN system, used by Thomson Reuters to measure the sentiment of asset-related media news, has already been utilized in various previous studies (Akyildirim et al., 2022; Rogmann et al., 2024; Smales, 2014; Maghyreh and Abdoh, 2020b; Nooijen and Broda, 2016; Shen et al., 2017). TRNA also clarifies whether a news item contains information that has already been released in the previous seven days. This data field helps us define novel news as news that does not contain any information released in the recent seven

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days, and old news as news items that do contain such information. Leveraging this classification, we measure daily novel and old news sentiment for each commodity aggregated from the sentiment of each news item.

First, we find that increases in both novel and old news sentiment are associated with higher returns in commodity futures. A one standard deviation increase in novel news sentiment corresponds to a 0.082% increase in returns, whereas a one standard deviation increase in old news sentiment corresponds to a larger 0.358% increase in returns. The stronger correlation between returns and sentiment in old news suggests that investors are more significantly influenced by news containing old information than by news with only new content. This finding aligns with the theory of investor inattention (Gilbert et al., 2012), indicating that investors may struggle to distinguish between old and new content in media news, leading to an overreaction to old news sentiment. When further examining the impact of positive and negative news sentiment, we find that commodity futures returns are more affected by negative news sentiment than positive news sentiment only for old news but not novel news.

Second, we find only old news sentiment triggers investor overreaction on the news day, resulting in statistically significant reversals in commodity futures returns over the five following trading days. This pattern aligns with the investor inattention theory (Gilbert et al., 2012), indicating that investors find it difficult to separate old and new information in a new piece of news and overreact to the pieces of old information. Additionally, the aggregated effects of novel and old news sentiment are similar after 30 trading days, suggesting that overreaction to old news sentiment is mostly reversed within 30 days and no longer significantly impacts returns beyond this period. We also find that this impact is more pronounced for first nearby contracts.

Finally, we find that commodity futures returns are more influenced by old news sentiment than by novel news sentiment during periods of high financial stress and uncertainty. We use the VIX index and financial stress indices from the US Government Office of Financial Research (OFR FSI) and the Federal Reserve Bank of St. Louis (STL FSI) to measure financial uncertainty and stress. These stress indices are commonly used in finance literature to evaluate the effects of financial stress (Cleary and Hossain, 2020; Converse et al., 2023; Della Corte et al., 2016; Gkillas et al., 2020; González-Uribeaga and Rubio, 2016; Stolbov et al., 2022). The findings indicate that, across all stress and uncertainty indices, the coefficients of interactive terms between each index and old news sentiment are higher in both magnitude and t-statistics compared to those between the same index and novel news sentiment. The results suggest that commodity futures returns are more responsive to old news sentiment than novel news sentiment under higher uncertainty and financial stress. The findings are consistent when using credit, equity, safe assets, funding, and volatility stress indices, the stress indices for the US and other advanced economies, and when using the lag of financial stress and uncertainty indices.

Our study contributes to the literature on media news sentiment in the commodity market (Abdollahi, 2023; Akyildirim et al., 2022; Li et al., 2021; Rogmann et al., 2024; Smales, 2015, 2014) by investigating the impact of both novel and old news sentiment on commodity futures returns. We demonstrate that these two types of sentiment trigger different response patterns in commodity futures returns. Additionally, our study contributes to the literature on the impact of novel and old news on asset returns (Fedyk and Hodson, 2023; Gilbert et al., 2012; Tetlock, 2011). Whereas Fedyk and Hodson (2023) and Tetlock (2011) investigate the impact of old news using different measures based on the volume of old information in news items, our focus is on the sentiment of old news and novel news. Our study highlights the significantly higher impact of old news sentiment compared to novel news sentiment, emphasizing the importance of old news sentiment in driving price movement. In our sample, we observe an overreaction of returns to old news sentiment but not to novel news sentiment. This suggests that the incorporation of old information and attached

sentiment in media news is a significant factor in driving commodity returns.

Last but not least, our study also contributes to the literature on the impact of sentiment under different market conditions. According to Birru and Young (2022), sentiment has a stronger impact during periods of higher uncertainty, both daily and yearly. García (2013) finds a significant increase in the effect of news sentiment during recession periods. In this study, we further focus on the impact of old and novel news sentiment and show that the intensified impact of news sentiment during times of higher uncertainty and stress is greater in magnitude and more significant for old news sentiment than for novel news sentiment. This emphasizes the impact of media news when the market is experiencing a higher level of financial stress and uncertainty, especially for the commodity futures market.

The rest of our paper is organized as follows. Section 2 reviews the literature. Section 3 describes the data source and variable measurement. Section 4 discusses the results of the study. Section 5 concludes. The appendix includes a description of the TRNA data source and tables about the key fields in this data.

2. Literature review

2.1. Media news and financial markets

The academic literature has thoroughly examined the link between media news and financial markets, with numerous studies providing strong evidence of this connection. One of the first studies on this topic was conducted by Niederhoffer (1971), which analyzes New York Times headlines and groups them by meaning and a multilevel good-bad scale. The study finds that stock prices tend to react strongly to news events, particularly negative ones, such as plane crashes that result in major losses for airlines. Subsequent studies focus more on the influence of media on financial markets, with Tetlock (2007) analyzing the negative tone in articles in the Wall Street Journal, while García (2013) analyzing the tone of New York Times reports. These studies consistently find that media tone can predict stock price movement.

The influence of media news on financial markets can be divided into two categories: market information provision and investor behavioral impact (Raimondo, 2019). When looking at media news from the perspective of providing information, it can help lower the level of asymmetric information in the market by spreading asset-related information, which, in turn, can enhance market efficiency. For example, Tetlock et al. (2008) find that a negative tone in news articles can convey information about a firm's fundamentals that are difficult to measure, which is then quickly incorporated into stock prices. On the other hand, when the media is viewed as a stimulus for investor reactions, previous studies focus on analyzing its impact on the financial market from various behavioral finance perspectives. These perspectives concentrate on the many aspects of behavioral biases and irrational trading (Ahern and Sosyura, 2015; Barber and Odean, 2008; Chan, 2003; Gurun and Butler, 2012; Hillert et al., 2018; Tetlock, 2007, 2011).

2.2. Sentiment in commodity markets

There is a wealth of research on the effect of investor sentiment on trading activities, returns, and volatility in the equity market (Baker and Wurgler, 2007; Ding et al., 2019; Du and Hu, 2020; Renault, 2017; Sun et al., 2016). Studies on the role of investor sentiment in the commodity market have also significantly increased mostly in the past decades (Bahloul and Bouri, 2016; Du and Zhao, 2017; Huang et al., 2018; Ji and Guo, 2015b,a; Maghyereh and Abdoh, 2020a; Maghyereh et al., 2020; Nguyen and Walther, 2020; Pan, 2018; Shahzad et al., 2017; Smales, 2014, 2015, 2017; Smales and Lucey, 2019). The same set of theories have been proposed to explain the impact of investor sentiment on both the commodity market and the equity market.

Accordingly, studies on investor sentiment in commodity markets also often seek support from behavioral finance theories.

Earlier studies have shown evidence of investor sentiment affecting specific commodity markets. These studies indicate that investor sentiment notably influences the price and volatility of both the oil market (Han et al., 2017; Ji et al., 2019; Li et al., 2019) and the gold market (Huang et al., 2019; Pan, 2018; Smales and Lucey, 2019). Gao and Süß (2015) examined market sentiment's impact on commodity futures returns and discovered that investor sentiment can predict commodity futures returns on a daily basis. Fernandez-Perez et al. (2020) suggest that attention to weather, disease, geopolitical, or economic threats, as measured by internet search volume, significantly affects the pricing of commodity futures. For a broader set of commodities, Bahloul and Bouri (2016) find that sentiment measured by the change of trader position can explain the return and volatility of major commodity futures markets.

Given the critical role of media news as a primary source of commodity information for investors, prior research has investigated how news sentiment, investor attention, and media coverage influence the dynamics of commodity futures markets (Li et al., 2019; Maghyereh and Abdoh, 2022a; Omura and Todorova, 2019; Smales, 2014, 2015, 2017). Smales (2014, 2015) examine the sentiment of gold-related news using the Thomson Reuters News Analytics data (TRNA), confirming that news sentiment significantly impacts the return and volatility of gold futures. Omura and Todorova (2019) explore the energy and metal futures markets, discovering an asymmetric influence of news sentiment on the futures prices of these commodities. Yang et al. (2019) analyze oil futures volatility and validate the predictive power of investor sentiment as measured by the FEARS index. Maghyereh et al. (2020) use the news sentiment measure for eight commodities in the energy and precious metals sectors from Thomson Reuters MarketPsi Indices (TRMI) data. Their results are consistent with the affect infusion theory (Forgas, 1995), which indicates that investors tend to lower their risk aversion when positive news sentiment is elevated, thereby affecting the price and volatility of commodity futures. Maghyereh and Abdoh (2022a) employ the market news sentiment measure from economic and financial news articles and show that this is a signal for price bubbles in some metal markets.

2.3. The impact of new and old media news

Regarding the impacts of novel and old media news on asset prices, Huberman and Regev (2001) examine the effect of an old news item in the New York Times. Although this information was previously published in the Nature Journal, the repost still triggered a reaction that increased the stock price to more than three times the last close. Tetlock (2011) argues that individual investors become confused between new and old information, causing them to react irrationally to old information. This study indicates that individual investors overreact to old information, which subsequently causes returns to reverse. This conclusion is consistent with the sentiment effect but cannot be explained by the fundamental value of information.

There are two behavioral theories that can explain the effect of old news. The first is the theory of bounded rationality (Simon, 1955; Shiller, 2003; Simon, 1997). This theory argues that human decision-making is restricted by their limitations in information, cognition, and time. Merton (1987) develops a model where investors in the market have limited information about particular assets and act solely based on their information. Hong and Stein (1999) argue that asset information is diffused gradually across investors, and thus, returns underreact to news in the short run. A second possible explanation for the influence of old news is the theory of investor inattention (Gilbert et al., 2012). This theory suggests that when old information is published again in the media news, investors confuse old information with new information, triggering an overreaction of asset returns to old news. Fedyk and Hodson (2014) examine news items that contain a combination of old and new information and show that investors are strongly influenced by old news items with complex interconnected structures.

Table 1

Descriptive statistics of commodity futures returns.

This table reports the basic descriptive statistics on the returns of the futures contracts in the sample. The default unit of return is 1%. Descriptive statistics are calculated for daily frequency. Each futures contract selected in the sample is the most recent contract whose expiration date occurs no earlier than 60 days after the selection date.

Commodity name	Mean	St. dev	Min	Max
Corn	0.004	1.665	-7.802	9.048
Cocoa	0.008	1.741	-9.480	8.653
Crude oil	0.030	2.273	-28.881	22.161
Cotton	0.006	1.586	-7.093	7.179
Gold	0.030	1.116	-9.345	8.987
Copper	0.045	1.712	-11.050	12.349
Coffee	-0.010	1.923	-10.644	12.828
Natural gas	-0.042	2.566	-18.302	18.686
Orange juice	0.009	1.862	-12.273	16.273
Gasoline	0.055	2.179	-22.079	17.142
Soybean	0.039	1.436	-7.032	6.644
Sugar	0.014	1.813	-11.632	8.934
Wheat	-0.011	1.916	-9.530	9.101
Total	0.014	1.864	-28.881	22.161

3. Data and methodology

3.1. Data on commodity futures markets

We use commodity futures price data from www.barchart.com, a reliable and commonly used data source in previous studies on commodity markets (Boons and Prado, 2019; Szymanowska et al., 2014). The data provides historical information on the prices of various futures contracts, making it convenient and accurate for selecting and monitoring price movements. This study uses data on 13 commodities, representing four types: three energy commodities (Crude oil, Gasoline, and Natural gas), three grain commodities (Corn, Soybean, and Wheat), two metal commodities (Gold, Copper), and five soft commodities (Cocoa, Coffee, Cotton, Sugar, and Orange juice). These commodities' futures contracts are traded on reputable exchanges such as NYMEX, CBOT, COMEX, and ICE. Detailed information on the futures contracts and their specifications is presented in Table A.3 in Appendix.

This study's sample period is from January 2003 to March 2021, aligned with the availability of media news data. Thirteen commodities are selected based on the availability of media news data before 2010. For each day and commodity, we follow the approach of Boons and Prado (2019) by using the most recent nearby contracts with at least 60 days remaining until their last trading day. This selection criterion helps avoid abnormal price movements as contracts approach their expiration date (Szymanowska et al., 2014). Following Boons and Prado (2019), we calculate the excess returns on fully collateralized futures positions as follows.

$$r_{i,t+1} = \frac{F_{i,t+1}}{F_{i,t}} - 1 \quad (1)$$

where $R_{i,t+1}$ is the return of the first nearby futures contract of commodity i at the day t . $F_{i,t}$ is the price of the first nearby futures contract of commodity i at the day t . Table 1 presents the main descriptive statistics for returns of the selected commodity futures contracts. The futures contracts' average returns are close to zero, and each commodity's maximum and minimum return values are relatively similar in magnitude. The descriptive statistics also reveal that futures contracts for energy commodities have greater volatility compared to other commodities. Furthermore, Gold consistently demonstrates the highest returns over time.

3.2. Data on media news sentiment

This study uses unique data from Thomson Reuters, specifically Thomson Reuters News Analytics (TRNA) provided by Thomson Reuters. TRNA compiles news from Reuters and other regulatory news

services to deliver timely updates on equities and commodities. Our study focuses on sentiment data and the relevance of news to commodities. We specifically select 13 commodities with historical futures price records from before 2010 until 2021.

TRNA data provides insights into the relevance of news for each commodity. Relevance in TRNA is measured based on the percentage of content related to a specific commodity in a news story. If the commodity's name is mentioned in the news title, the relevance is set at 100%. Additionally, if the news discusses a commodity group to which the commodity belongs without directly mentioning its name, the relevance to commodities in that group is also set at 100%. For example, if a news item provides general information about the energy supply without explicitly mentioning the names of energy commodities, the relevance to crude oil, natural gas, or gasoline is considered 100%. To ensure a focus on news items that are closely related to each asset, our study only selects news items that are 100% relevant and directly mention the name of the commodity.

TRNA data provides information on the number of news items that are related to a given news item within different time periods (12 h, 24 h, three days, five days, and seven days) prior to its publication. Based on this information, we classify a news item as “novel news” if it is unrelated to any news released in the past seven days. Within our news sample, we distinguish between novel and old newsgroups. To examine the impact of novel and old news sentiment, we first measure the sentiment of news articles from our targeted new groups. TRNA provides probabilities of positive, negative, or neutral messages and categorizes sentiment based on the highest probability. The sentiment determination is stored in the “SentimentClass” field. We have checked with TRNA, and the data provider confirmed that all updates to the TRNA sentiment dictionary are applied from the time of the update and do not affect the historical TRNA sentiment data. This further minimizes any look-ahead bias in the data.

Our approach to measuring news sentiment aligns with Smales (2014), relating sentiment to the dominant aspect of news tone. If a news item has a neutral sentiment class, its sentiment is assigned as 0. The key fields in TRNA data are described in Appendix.

$$Sen_{c,i}^{news} = \begin{cases} P_{c,i}^{pos} & \text{if the sentiment class takes the value of 1} \\ -P_{c,i}^{neg} & \text{if the sentiment class takes the value of -1} \\ 0 & \text{if the sentiment class is 0} \end{cases} \quad (2)$$

where $Sen_{c,i}^{news}$ is the sentiment of commodity c in the news i . $P_{c,i}^{pos}$ and $P_{c,i}^{neg}$ are the probability that the news i delivers positive and negative aspects of tone on the commodity c , respectively.

The daily news sentiment of a specific commodity is determined by aggregating the sentiment of each news item selected for that day. Only the news released within a 24-h period before the daily settlement time of each commodity futures is used to calculate the daily news sentiment. Considering that the news presented by the system is based on the Coordinated Universal Time (UTC) zone, the timestamps in the TRNA data are adjusted to the Central Standard Time (CST) zone for consistency and accuracy. This study adopts the sentiment measure that Smales (2014) proposed and uses this measure for all news (full news sample), novel news, and old news. Specifically, the daily news sentiment for a commodity c on the day t can be calculated as the sum of positive news (n_{pos}), negative news (n_{neg}), and neutral news (n_{neu}). We calculate daily news sentiment for all news (full news sample) and for the subsamples of novel news and old news.

$$Sen_{c,t} = \frac{\sum_{i \in pos_news} 1 \times P_{c,i}^{pos} + \sum_{j \in neg_news} (-1) \times P_{c,j}^{neg}}{n_{pos} + n_{neu} + n_{neg}} \quad (3)$$

Table 2 presents the main descriptive statistics for the sentiment of all news, novel news, and old news. The table shows that the average news sentiment is different between novel news and old news for each commodity. The average novel news sentiment for all commodities is around -0.010 , whereas the average old news sentiment is -0.027 .

Table 2

Descriptive statistics of news sentiment.

This table provides key descriptive statistics about news sentiment for each commodity. The results are shown for three categories of news: all news, novel news, and old news. The news in the full sample are those with 100% relevance to a specific commodity and directly mentioning the commodity's name. Novel news includes news items that have content unrelated to any content in the previous seven days. Old news consists of items containing information released in the previous seven days.

Commodity		All news	Novel news	Old news
Corn	Mean	0.077	0.018	0.084
	St. dev	0.298	0.353	0.303
Cocoa	Mean	-0.062	-0.055	-0.056
	St. dev	0.307	0.413	0.333
Crude oil	Mean	-0.048	-0.002	-0.063
	St. dev	0.150	0.161	0.174
Cotton	Mean	-0.156	0.007	-0.193
	St. dev	0.245	0.385	0.252
Gold	Mean	0.025	0.052	0.021
	St. dev	0.209	0.306	0.225
Copper	Mean	-0.060	-0.068	-0.060
	St. dev	0.337	0.399	0.376
Coffee	Mean	-0.047	-0.031	-0.050
	St. dev	0.285	0.400	0.318
Natural gas	Mean	-0.020	0.056	-0.046
	St. dev	0.228	0.292	0.274
Orange juice	Mean	0.029	-0.050	0.045
	St. dev	0.351	0.250	0.355
Gasoline	Mean	-0.049	-0.022	-0.059
	St. dev	0.314	0.404	0.360
Soybean	Mean	0.050	0.015	0.060
	St. dev	0.277	0.348	0.289
Sugar	Mean	-0.059	-0.052	-0.055
	St. dev	0.272	0.371	0.303
Wheat	Mean	0.007	-0.025	0.012
	St. dev	0.215	0.330	0.228
All	Mean	-0.023	-0.010	-0.027
	St. dev	0.279	0.346	0.303

Also, in our sample, the correlation between novel and old news sentiment is approximately 0.189 across all commodities. Crude oil has the highest correlation at 0.358, while most other commodities show correlations below 0.2. This weak correlation indicates that old news sentiment has some distinct aspects compared to novel news sentiment.

Previous studies have also examined the separate effects of positive and negative news or focused on specific aspects of news tone (García, 2013; Smales, 2014; Tetlock, 2007). We calculate positive and negative news sentiment for each commodity to differentiate between the impact of positive and negative news.

$$Sen_i^{pos} = \frac{\sum_{i \in pos_news} 1 \times P_i^{pos}}{n_{pos} + n_{neu} + n_{neg}} \quad (4)$$

And

$$Sen_i^{neg} = \frac{\sum_{j \in neg_news} (-1) \times P_j^{neg}}{n_{pos} + n_{neu} + n_{neg}} \quad (5)$$

3.3. Data on control variables

This study uses control variables sourced from Datastream, WRDS, and the St. Louis Federal Reserve Economic Data (FRED). Previous research has extensively examined the impact of macro variables on commodity returns (Caporale et al., 2017; Elder et al., 2012; Hamadi et al., 2017; Liu et al., 2018; Roache and Rossi, 2010; Smales, 2017; Yin and Han, 2016). This study follows the literature by including three daily macro variables: the TED spread, the US Dollar index, and the one-month T-bill, as well as two exchange-related variables, the VIX and the stock market return. The stock market return is calculated as the value-weighted return of the portfolio consisting of all US stocks in the CRSP database. A detailed list of these macroeconomic variables is presented in Table 3, and their descriptive statistics are presented in

Table 3

Macroeconomics variables and corresponding literature.

This table reports the macro variables used as control variables in this study. These macro variables are collected from Datastream, WRDS, and the St. Louis Federal Reserve Economic Data (FRED) at daily frequency. In addition, this table also annotates previous studies that used these macro variables to account for the returns on financial assets.

Macro variable	Frequency	Literature
TED spread	Daily	Büyüksahin and Robe (2014), Koch (2014), Prokopczuk et al. (2019)
US Dollar index	Daily	Akram (2009), Nazlioglu and Soytaş (2012), Yoon et al. (2019)
One-month T-bills	Daily	Campbell (1987), Fama and French (1989), Hong and Yogo (2012)
Stock market return	Daily	Mensi et al. (2013), Yoon et al. (2019), Balcilar et al. (2021)
VIX	Daily	Zhang et al. (2017), Bahloul et al. (2018), Hu et al. (2020)

Table 4

Descriptive statistics of daily macro and stock-market-related variables.

The table provides the basic descriptive statistics of the macroscopic variables used in the study and the units of these variables. Statistics for the daily variables were calculated on a sample from 1 January 2003 to 31 March 2021.

Variable	Unit	Mean	Std. Dev.	Min	Max
TED spread	Percentage	0.417	0.417	0.090	4.580
US Dollar index	Point index	87.493	7.936	71.330	103.290
One-month T-bills	Percentage	0.005	0.006	0.000	0.022
Stock market return	Percentage	0.057	1.203	-11.994	11.354
VIX	Point index	19.073	9.000	9.140	82.609

Table 4. Because the US Dollar index is non-stationary, we use the first difference form of the US Dollar index in the regression analysis.

Table 5 presents correlations between key variables in the study. The correlation between old news sentiment and all news sentiment is approximately 0.206, which is higher than the correlation between novel news sentiment and all news sentiment (0.082). Additionally, both novel and old news sentiment show very low correlations with other control variables. These findings suggest that news sentiment may provide distinct information compared to macro and stock market variables.

4. Analysis and discussion

4.1. Contemporaneous news sentiment and commodity futures return

This section examines the relationship between media news sentiment and commodity futures returns. Unlike previous studies that used newspapers (García, 2013; Tetlock, 2007), this study focuses on media news. Media news has become a popular and timely source of information due to the growth of media outlets and internet access. In Tetlock (2007) and García (2013), newspaper-based sentiment is calculated for the newspaper articles published early in the morning, and the content of these news items is prepared almost the day before. Therefore, market movements on the same day could not impact the sentiment of the early morning news. Unlike newspapers, media news is released during the day, often during market opening hours. Moreover, many commodity futures trade for up to 23 h a day, so news items may contain information and sentiment that impact asset prices within the trading day.

There is a concern about endogeneity, where same-day news sentiment may also be influenced by returns. To address this, we adopt the argument by Smales (2014) that news sentiment is minimally affected by concurrent futures returns. Smales (2014) randomly examines news items for each commodity and finds that same-day returns are rarely mentioned. This is expected since closing prices are only known at settlement time, whereas news sentiment for the same trading day is calculated based on a 24-h period before settlement. Additionally, we control for past returns and relevant macroeconomic and stock

market variables to account for other factors that may affect commodity futures returns and potentially influence same-day news sentiment. Furthermore, because news sentiment is autocorrelated and potentially impacts returns, omitting contemporaneous news sentiment when modeling returns would result in significantly biased estimates of the lagged news sentiment's impact. Amihud et al. (2009) and Stambaugh (1999) provide a detailed explanation of the importance of including the contemporaneous explanatory variables when they are autocorrelated. Also, only a small portion of news items are released close to the daily settlement time in a 24-h period. For example, we find that, on average, only 10% of news items are released across all commodities up to two hours before the daily settlement time.¹

This study first investigates the relationship between commodity futures returns and contemporaneous news sentiment. Whereas news sentiment is measured using individual commodity news, it may potentially be correlated with macro information and stock market movements. To account for this, the regression model includes stock market return and VIX as controls for the stock market impact and the US Dollar index, TED spread, and one-month T-bill rate as macro control variables. We control for year-fixed effects and commodity-fixed effects. We also control for the weekday effect, using day-of-the-week dummies. The estimated pooled regression is as follows:

$$r_{i,t} = \alpha + \beta Sent_{i,t} + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (6)$$

where $r_{i,t}$ is the return of commodity i on the day t . $Sent_{i,t}$ is the net news sentiment of commodity i aggregated for the 24-h period preceding the daily settlement time of day t . $Contr_{q,t}$ is the value of control variable q at time t . $F_{i,t}$ are the fixed effects controlled in the model. Following Boons and Prado (2019), due to the low autocorrelation of returns for each commodity futures, we adjust the model using clustered standard error in the time dimension. The news sentiment variables are all standardized. Table 6 reports the estimated results.

In Table 6, the correlation between contemporaneous news sentiment and returns is statistically significant for all news news sentiment. A one standard deviation increase in news sentiment corresponds to a 0.37% increase (t-stat = 15.24) in returns. When incorporating macroeconomic and stock market variables in Model 2, the coefficient for all news sentiment decreases slightly from 0.41 to 0.37, with the t-statistics remaining mostly unchanged. This suggests that the correlation between returns and news sentiment is weakly influenced by macroeconomic and stock market factors. Our result aligns with other research on news sentiment (Tetlock, 2007; García, 2013; Smales, 2014) and research on investor sentiment's impact on the commodity market, including trader position-based sentiment (Bahloul and Bouri, 2016), news-based economic and consumer-based sentiments (Maghyereh and Abdoh, 2022a), and macro events such as the global financial crisis (Maghyereh and Abdoh, 2022b; Shahzad et al., 2017) and COVID-19 pandemic (Maghyereh and Abdoh, 2022b).

Model 3 only regresses the return of commodity futures on novel news sentiment and lag of return. In contrast, Model 4 uses a similar model on old news sentiment instead of novel news sentiment. Model 5 includes both novel and old news sentiment, whereas Model 6 adds extra control variables and fixed effects. All these four models confirm that increases in both novel and old news sentiment are associated with higher returns. Specifically, one standard deviation increase in novel news sentiment and old news sentiment correspond to a 0.08% (t-stat = 8.97) increase and 0.36% (t-stat = 14.60) increase in commodities futures returns, respectively.² We report the test for the difference between the impact of old and novel news sentiment in the last two

¹ The maximum proportion of news released up to two hours before daily settlement time is 16.6% for orange juice. Also, an average of 4.4% of news items are released up to one hour before daily settlement time, with gasoline having the maximum at 7.4%.

Table 5

Correlation between variables.

This table presents correlations among key variables examined in the study. The research sample spans from January 2003 to March 2021.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Return	(1)	1.000							
All news sentiment	(2)	0.205	1.000						
Novel news sentiment	(3)	0.082	0.474	1.000					
Old news sentiment	(4)	0.206	0.898	0.189	1.000				
CRSP return	(5)	0.167	0.057	0.015	0.057	1.000			
Risk-free rate	(6)	−0.001	0.052	0.041	0.042	−0.017	1.000		
VIX	(7)	−0.040	−0.037	−0.022	−0.032	−0.123	−0.223	1.000	
TED spread	(8)	−0.022	−0.014	0.001	−0.017	−0.034	0.267	0.504	1.000
D Usdollar	(9)	−0.173	−0.069	−0.022	−0.071	−0.120	−0.014	0.031	0.014

Table 6

Contemporaneous news sentiment and Commodity futures return.

This table presents the estimated results of the pooled regression models that examine the relationship between commodity futures returns and news sentiment. The results are shown for three categories of news: all news, novel news, and old news. Novel news includes news items that have content unrelated to any content in the previous seven days. Old news consists of items containing information released in the previous seven days. The first two models display the results for all news sentiment without and with macro and stock market control variables. The third and fourth models show the results for novel news sentiment and old news sentiment, respectively, without control variables. The fifth and sixth models incorporate both novel and old news sentiment. All models include controls for commodity fixed effect, year fixed effect, and day-of-week fixed effect. For models 5 and 6, we test the difference between the impact of old and novel news sentiment. The test is reported in the last two rows of the table. “Old - Novel news sentiment” is the difference between coefficients of old and novel news sentiment, while t-stat shows the *t*-statistics of the test for whether this difference is statistically different from 0. The coefficients are clustered by time dimension.

Model	1	2	3	4	5	6
Variables	ret (t)	ret (t)	ret (t)	ret (t)	ret (t)	ret (t)
All news sentiment (t)	0.406*** (13.265)	0.370*** (15.244)				
Novel news sentiment (t)			0.151*** (10.858)		0.086*** (8.603)	0.082*** (8.970)
Old news sentiment (t)				0.408*** (13.193)	0.393*** (13.112)	0.358*** (14.600)
Return (t−1)	−0.068*** (−11.343)	−0.066*** (−10.902)	−0.023*** (−3.361)	−0.070*** (−11.322)	−0.072*** (−11.636)	−0.069*** (−11.092)
CRSP return (t)		0.204*** (8.478)				0.204*** (8.522)
Risk-free rate (t)		−21.223** (−2.489)				−20.725** (−2.418)
VIX (t)		−0.003* (−1.790)				−0.003* (−1.764)
TED spread (t)		−0.084 (−0.779)				−0.081 (−0.731)
Δ Usdollar (t)		−0.631*** (−6.035)				−0.628*** (−5.979)
Constant	0.034 (1.440)	0.148** (2.288)	0.023 (1.411)	0.043 (1.608)	0.040 (1.514)	0.151** (2.248)
Commodity and Time fixed effects	YES	YES	YES	YES	YES	YES
SE clustered by time	YES	YES	YES	YES	YES	YES
Observations	57,729	57,729	57,729	57,729	57,729	57,729
R-squared	0.047	0.092	0.008	0.048	0.050	0.094
Old - Novel news sentiment					0.307	0.276
t-stat					11.022	11.076

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5%, and 10% levels, respectively.

rows of Table 6. For Model 6, which includes all control variables, the difference between the impact of old news sentiment and novel news sentiment is 0.276, with a *t*-statistic of 11.076. The result implies that return is more sensitive to old news than novel news sentiment.

In Table 6, returns are more correlated with the sentiment of old news rather than novel news sentiment. If the market is efficient and

news sentiment merely reflects fundamental information, the correlation of return with novel and old news should be similar since old information should not affect returns. Hence, the result implies that investors respond differently to the sentiment of old news and novel news. The R-squared values in Model 3 and Model 4 indicate that old news sentiment explains a larger portion of the variance in commodity futures returns compared to novel news sentiment. This finding aligns with the theory of investor inattention (Gilbert et al., 2012), which suggests that investors find it difficult to differentiate between old and new information in news items. As a result, they may overreact to old news.

Control variables provide insights into the relationship between commodity markets, equity markets, and macroeconomic factors. Table 6 indicates that higher stock market returns correspond to higher commodity futures returns, suggesting that commodity futures may not serve as an effective hedge for equity markets. In terms of investor

² In this analysis, we correct standard errors by clustering using the time dimension. We chose this approach due to the weak autocorrelation of commodity futures returns for each commodity. Table A.4 in the Appendix presents the results of a robustness test where we cluster using the commodity dimension and both commodity and time dimensions. The results show that our findings in Table 6 remain statistically significant across different clustering settings.

sentiment, a higher stock market return and a lower VIX suggest increased investor sentiment in the stock market. However, the positive correlation between stock market returns and the negative correlation between the VIX and commodity futures returns indicate a partial synchronization of investor sentiment between the commodity futures and equity markets. A higher T-Bill rate may signal increased inflation, credit, and liquidity risks in the capital market. Table 6 reveals a negative correlation between the T-Bill rate and commodity futures returns, suggesting that investor sentiment in the commodity futures market is influenced by macroeconomic and capital market sentiment. The negative correlation between returns and changes in the US Dollar Index reflects the impact of the US Dollar on commodity market transactions, as most commodity futures are denominated in US Dollars. An increase in the value of a currency relative to other currencies corresponds to an adjustment in commodity prices.³

To further address potential endogeneity issues in our analysis, we apply an instrumental variables (IV) approach. We use the lagged market sentiment as an instrumental variable, measuring market sentiment as the average sentiment across all commodities. This equal-weighted average captures macro shocks affecting all commodities' sentiment. Additionally, as market sentiment might reflect macro shocks influencing same-day commodity futures returns, we include contemporaneous market sentiment in the return model. The use of macro or group variables as instrumental variables to capture exogenous shocks is common in the literature (Beck et al., 2014; Cole et al., 2014; Ferrell et al., 2016; Laeven and Levine, 2009; Lin, 2016; Lin et al., 2011; Ma et al., 2019). Our selection of market sentiment as an instrumental variable follows the same principle. Table 7 presents the results of IV regression for all news, novel news, and old news. For each set of news, the first column shows the results for the first stage of the 2SLS regression, and the second column shows the results for the second stage.

The results show that the lagged market sentiment significantly predicts commodity-specific news sentiment in the first stage of the IV regression for all news types—novel, old, and all news. In the second stage, the predicted news sentiment from the first stage significantly correlates with commodity futures returns across all news categories. The Kleibergen–Paap rk LM statistic and Cragg–Donald Wald F statistic indicate that the selected instrument is both valid and not weak. These findings suggest that contemporaneous news sentiment impacts commodity futures returns.

4.2. Positive and negative news sentiment

In this section, we investigate the relationship between commodity futures returns and news sentiment (both novel and old) and further explore the impact of positive and negative news sentiment. García (2013) and Smales (2014) have documented that asset returns are positively affected by positive news sentiment and negatively affected by negative news sentiment. Following the tests in these studies, we run the following model:

$$r_{i,t} = \alpha + \beta_1 Sent_{i,t}^{pos} + \beta_2 Sent_{i,t}^{neg} + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (7)$$

where $Sent_{i,t}^{pos}$ and $Sent_{i,t}^{neg}$ are positive and negative news sentiment for the commodity i on the day t , respectively. The estimated results are reported in Table 8.

³ The literature has tested several fundamental factors that can predict commodity futures returns, including liquidity and basis (Szymanowska et al., 2014), momentum (Bakshi et al., 2017), basis momentum and volatility (Boons and Prado, 2019). These factors are typically used as return predictors at weekly and monthly frequencies. We measure these factors at a daily frequency and report the results in Table A.5 in Appendix. Notably, even when controlling for these factors, the magnitude and t-statistics of the news sentiment coefficients change very little. This suggests that the impact of news sentiment is only weakly linked to these fundamental factors.

Models 1 and 2 in Table 8 show that positive news sentiment is associated with an increase of 0.21% (t-stat = 13.04) in returns, whereas negative news sentiment is associated with a decrease of 0.23% (t-stat = -11.46) in returns. In both Model 3 and Model 4, the impact of negative novel news sentiment is slightly lower than that of positive novel news sentiment. In Model 4, which includes all control variables, a one standard deviation increase in positive and negative novel news sentiment is associated with a 0.091% increase and a 0.086% decrease in return, respectively. We report the test for the difference between the impact of positive and negative news sentiment in the last two rows of Table 8. For novel news, the difference between the magnitude of coefficients of positive and negative novel news sentiment in Model 4 is minimal, only 0.004, with a t-statistic of 0.255. This indicates that there is no statistically significant difference between the impact of positive and negative novel news sentiment.

Models 5 and 6 of Table 8 present the impacts of positive and negative old news sentiment. In Model 6, a one standard deviation increase in positive old news sentiment is associated with a 0.202% increase in return, whereas a one standard deviation increase in negative old news sentiment is associated with a 0.244% decrease in return. The difference between the magnitude of coefficients of positive and negative novel news sentiment in Model 6 is 0.042, which is statistically different from 0 with a t-stat of 2.431. The difference between the magnitudes of coefficients for positive and negative old news sentiment in Model 6 is 0.042, which is statistically significant with a t-statistic of 2.431. This result indicates that, only for old news, returns react more strongly to negative news sentiment than positive news sentiment. This finding suggests that only the sentiment of old news rather than novel news triggers an asymmetric impact on returns, aligning with loss aversion theory (Tversky and Kahneman, 1991).

4.3. Lagged effects of news sentiment

In this section, the study analyzes the lagged effect of news sentiment on commodity futures returns. Media news differs from newspapers in that media news is published throughout the day. As a result, the information and additional sentiment content in the news have the potential to impact returns on the same day. The above analysis has also shown highly statistically significant correlations between contemporaneous news sentiment and commodity futures returns. Therefore, models to assess the lagged effects of news sentiment require the presence of contemporaneous news sentiment. Amihud et al. (2009) and Stambaugh (1999) note that when an independent variable affects the dependent variable contemporaneously and the independent variable is autocorrelated, omitting the contemporaneous term leads to biased estimations of the effects of the lagged independent variable. For this reason, the lagged impact of news sentiment on returns is estimated using the following model:

$$r_{i,t} = \alpha + \beta_0 Sent_{i,t} + \sum_{j=1}^k \beta_k Sent_{i,t-j} + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (8)$$

For the first set of regressions, this study tests the lagged effect of the last five trading days. To distinguish the effects of novel news sentiment and old news sentiment, we run the models separately for novel news sentiment and old news sentiment. Table 9 shows the results of these models.

Table 9 supports the results in Table 6 that returns are more strongly correlated with the contemporaneous sentiment of old news than that of novel news. Adding the lag terms of news sentiment into the model, the R-squared of the novel news sentiment model presents a smaller value than that of the old news sentiment model, again indicating that old information plays a significant role in driving commodity futures returns. Furthermore, Table 9 reveals a striking contrast between the effects of novel news sentiment and old news sentiment. The estimated impact of the first five lags of novel news sentiment is minimal compared to the contemporaneous correlation. Additionally, none of the

Table 7

Robustness check: Instrument variable regressions.

This table reports the estimated results of the instrumental variable regression of commodity futures returns on novel and old news sentiment. We use lagged market sentiment as an instrumental variable for commodity-specific news sentiment. The lagged market sentiment is measured as the average sentiment across all commodities. This equal-weighted average captures macro shocks affecting all commodities' sentiment. The results are displayed for three news groups: all news, novel news, and old news. For each group of news, we report the estimated results of two stages of instrumental variable regression in two columns. The first column shows the first stage, where news sentiment is regressed on the lagged market sentiment and controlled for other variables in the second stage. The second column reports the results of the second stage where commodity futures return is regressed on the predicted news sentiment from the first stage. The standard errors of coefficients are clustered using the time dimension.

	All news		Novel news		Old news	
VARIABLES	Sent (t)	Return (t)	Sent (t)	Return (t)	Sent (t)	Return (t)
Sentiment (t)		3.228*** (6.301)		3.143*** (3.795)		6.380*** (3.473)
Market sentiment (t-1)	-0.065*** (-7.604)		-0.067*** (-7.966)		-0.033** (-2.573)	
Market sentiment (t)	0.925*** (91.781)	-2.113*** (-4.492)	0.427*** (30.873)	-0.468 (-1.515)	0.845*** (67.842)	-4.517*** (-2.805)
Return (t-1)	0.092*** (11.479)	-0.343*** (-9.487)	0.024*** (5.459)	-0.121*** (-7.679)	0.098*** (11.497)	-0.673*** (-4.029)
CRSP return (t)	0.005*** (4.803)	0.170*** (7.201)	-0.008** (-2.465)	0.212*** (7.467)	0.009*** (7.524)	0.127*** (5.237)
Risk-free rate (t)	1.216 (1.512)	-21.001** (-2.354)	-0.221 (-0.044)	-16.380 (-1.309)	0.023 (0.011)	-17.219 (-0.946)
VIX (t)	-0.000 (-0.972)	0.002 (0.715)	-0.002* (-1.787)	0.007 (1.457)	0.000 (0.450)	-0.000 (-0.099)
TED spread (t)	0.001 (0.090)	-0.097 (-0.731)	0.007 (0.304)	-0.116 (-1.079)	-0.011 (-0.507)	-0.025 (-0.109)
Δ Usdollar (t)	-0.002 (-0.381)	-0.559*** (-7.434)	0.019*** (2.790)	-0.624*** (-6.300)	-0.016*** (-3.297)	-0.463*** (-5.251)
Constant	-0.003 (-0.058)	0.063 (0.372)	0.077** (2.210)	-0.191* (-1.902)	-0.031 (-0.497)	0.251 (0.620)
Kleibergen-Paap rk LM statistic	12.642		13.677		4.738	
p-value	0.000		0.000		0.030	
Cragg-Donald Wald F statistic	33.525		29.977		8.455	

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5%, and 10% levels, respectively.

Table 8

The impact of positive and negative news sentiment on commodity futures return.

This table reports the estimated results of the pooled regression models of commodity futures returns on positive and negative news sentiment. The results are displayed for three new groups: all news news, novel news, and old news. Novel news includes news items that have content unrelated to any content in the previous seven days. Old news consists of items containing information released in the previous seven days. For each new group, this study conducts two model specifications. The first model only regresses returns on contemporaneous positive and negative news sentiment and the first lag of the return. The second model adds control variables, commodity fixed effect, year fixed effect, and day-of-week fixed effect into the first model. The standard errors of coefficients are clustered by time dimension.

	All news		Novel news		Old news	
Model	1	2	3	4	5	6
Variable	ret (t)	ret (t)	ret (t)	ret (t)	ret (t)	ret (t)
Positive news sentiment (t)	0.221*** (12.205)	0.206*** (13.035)	0.100*** (8.545)	0.091*** (8.906)	0.218*** (11.318)	0.202*** (12.097)
Negative news sentiment (t)	-0.259*** (-9.964)	-0.233*** (-11.463)	-0.091*** (-6.970)	-0.086*** (-7.283)	-0.270*** (-11.195)	-0.244*** (-13.302)
Return (t-1)	-0.071*** (-11.505)	-0.068*** (-11.174)	-0.023*** (-3.387)	-0.025*** (-3.737)	-0.073*** (-11.533)	-0.071*** (-11.149)
CRSP return (t)		0.204*** (8.507)		0.216*** (8.655)		0.203*** (8.544)
Risk-free rate (t)		-21.027** (-2.472)		-23.939** (-2.609)		-20.589** (-2.324)
VIX (t)		-0.003 (-1.602)		-0.006*** (-3.556)		-0.003* (-1.737)
TED spread (t)		-0.083 (-0.765)		-0.077 (-0.694)		-0.078 (-0.703)
Δ USdollar (t)		-0.628*** (-6.071)		-0.673*** (-5.833)		-0.626*** (-6.009)
Constant	0.020 (0.900)	0.129* (1.997)	0.021 (1.299)	0.213*** (3.034)	0.024 (0.958)	0.137* (2.000)
Observations	57,729	57,729	57,729	57,729	57,729	57,729
R-squared	0.050	0.094	0.008	0.060	0.050	0.094
Neg. - Pos. news sentiment t-stat	0.037 1.574	0.028 1.332	0.008 0.478	0.004 0.255	0.051 2.478	0.042 2.431

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5%, and 10% levels, respectively.

Table 9

News sentiment and Lagged impact on commodity futures return.

This table reports the estimated results of the pooled regression models of commodity futures returns on news sentiment and the first five lags of news sentiment. The results are displayed for three new groups: all news, novel news, and old news. Novel news includes news items that have content unrelated to any content in the previous seven days. Old news consists of items containing information released in the previous seven days. This study conducts the regression for each new group, including the control variables, commodity fixed effect, year fixed effect, and day-of-week fixed effect. The standard errors of coefficients are clustered by the time dimension. For all models, we test the difference between the impact of positive and negative news sentiment. The test is reported in the last two rows of the table. “Neg. - Pos. news sentiment” is the difference between the magnitude of coefficients of positive and negative news sentiment, while t-stat shows the t-statistics of the test for whether this difference is statistically different from 0.

	All news	Novel news	old news
VARIABLES	ret (t)	ret (t)	ret (t)
News sentiment (t)	0.406*** (15.151)	0.140*** (12.172)	0.414*** (15.188)
News sentiment (t-1)	-0.050*** (-4.674)	0.003 (0.288)	-0.056*** (-6.297)
News sentiment (t-2)	-0.047*** (-3.772)	-0.003 (-0.328)	-0.050*** (-5.325)
News sentiment (t-3)	-0.023** (-2.292)	-0.002 (-0.238)	-0.024** (-2.455)
News sentiment (t-4)	-0.026*** (-3.317)	0.006 (1.076)	-0.030*** (-3.169)
News sentiment (t-5)	-0.034*** (-3.702)	0.003 (0.490)	-0.038*** (-3.770)
Return (t-1)	-0.066*** (-10.511)	-0.026*** (-3.631)	-0.068*** (-10.434)
R-squared	0.095	0.060	0.096

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5%, and 10% levels, respectively.

lags of novel sentiment have significant impacts. This result implies that the impact of the novel news sentiment is mostly concentrated on the same day of returns, and no significant reversals are seen over the next five consecutive trading days. This pattern may also suggest that the sentiment of novel news mainly reflects the value of new information, and this set of information is quickly incorporated into the price. Consequently, we do not see any reversals in the following days.

However, over the five following trading days, we observe statistically significant reversals in the impact of old news sentiment on commodity futures returns. All of the coefficients of the five old news sentiment lags are significant and negative, suggesting that investors overreact to the old news sentiment. Specifically, the price is corrected gradually, starting on the next trading day. No drifts are observed in the impact of old news sentiment. The observed pattern conforms to investor inattention theory (Gilbert et al., 2012) that investors are confused by the combined old information in a new piece of news and overreact to the pieces of old information. In summary, the empirical evidence in Table 9 suggests that the sentiment of novel news does not induce significant overreaction in the commodity futures market. However, the sentiment of old information in the news may trigger investor overreaction. We also extend our analysis to lag $t-30$ for all, novel and old news sentiment. Table 10 shows the aggregated effect for each five-trading-day period from day $t-1$ to day $t-30$.

The results also show no evidence for the reversals of contemporaneous correlation for novel news sentiment. In each group of a five-day period from $t-1$ to $t-30$, the aggregated impacts of novel news sentiment on returns are very small in magnitude, and all are not statistically significant. The total impact of novel news sentiment from day t to day $t-30$ is approximately 0.1% (t-stat = 3.54). These results suggest that novel news sentiment again mostly delivers the information value and triggers no overreaction in investor behaviors. In contrast, we observe the reversals of contemporaneous correlation in the model of old news sentiment. The aggregated impacts of old

Table 10

News sentiment and its reversal effect on commodity futures returns.

This table reports the aggregated impacts of news sentiment on commodity futures returns for each 5-day period from the day $t-1$ to the day $t-30$, and for the period from the day t to the day $t-30$. This study conducts the regression with contemporaneous news sentiment and its first 30 lags. The results are displayed for three new groups: all news, novel news, and old news. The total sentiment impact, the corresponding t-statistics, and significance are calculated from the Wald test.

	All news	Novel news	Old news
News sentiment (t)	0.419*** (15.066)	0.142*** (12.161)	0.425*** (14.992)
Total sentiment effect (t-5 to t-1)	-0.130*** (-6.463)	0.014 (0.886)	-0.150*** (-8.998)
Total sentiment effect (t-10 to t-6)	-0.055*** (-3.552)	-0.004 (-0.238)	-0.050*** (-3.641)
Total sentiment effect (t-15 to t-11)	-0.028 (-1.666)	-0.010 (-0.502)	-0.023 (-1.698)
Total sentiment effect (t-20 to t-16)	-0.040** (-2.735)	-0.026 (-1.565)	-0.043*** (-4.478)
Total sentiment effect (t-25 to t-21)	-0.044** (-2.394)	-0.019 (-1.205)	-0.045** (-2.774)
Total sentiment effect (t-30 to t-26)	-0.014 (-0.867)	-0.001 (-0.078)	-0.015 (-0.972)
Total sentiment effect (t-30 to t)	0.108*** (4.958)	0.096*** (3.538)	0.100*** (4.479)

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5% and 10% levels, respectively.

news sentiment are all negative and statistically significant for most 5-day periods from day $t-1$ to day $t-25$. The total effect of old news sentiment from day t to day $t-30$ is 0.1% (t-stat = 4.48).

It is noted that old news comprises both novel information and old information. Hence, the almost equal total effect of novel news sentiment and old news sentiment from day t to day $t-30$ implies that investors' overreaction to old information sentiment in old news is mainly reserved within 30 trading days. Moreover, the effect pattern of all news sentiment is very similar to that of old news sentiment. After 30 trading days, the total impacts of both novel and old news sentiment stay around 10%. The result again supports the investor inattention theory (Gilbert et al., 2012) and provides additional evidence for the impact of old news (Tetlock, 2011) but from a different perspective, focusing on sentiment.

To check the robustness, we also examine the impact pattern of the lagged positive and negative news sentiment for both novel and old news. We conduct the same models as in Tables 9 and 10 but replace net news sentiment with positive and negative news sentiment. More specifically, the regressions in Table 11 follow the below functional form:

$$r_{i,t} = \alpha + \sum_{j=0}^k \beta_j^{pos} Sent_{i,t-j} + \sum_{j=0}^k \beta_j^{neg} Sent_{i,t-j} + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (9)$$

In Table 11, the first column reports the coefficients of contemporaneous positive news sentiment and its first five lags, and the second column reports similar coefficients for negative news sentiment. Table 11 also suggests no reversals of contemporaneous correlation for novel news sentiment, implying that investors adjust the price quickly following both positive and negative novel news sentiment. For old news, the reversal is observed strongly after a day for negative old news sentiment but not on the next three days from day $t-2$ to day $t-4$ and significant again on day $t-5$. In contrast, we observe that the reversal happens significantly on the following five days except for day $t-4$ for positive old news sentiment. This result implies that the price-correcting process for overreaction is more spread out for positive old news sentiment than for negative old news sentiment. To better capture this price reversal process, I conduct the model with 30 lags of

Table 11

The lagged impact of positive and negative news sentiment on commodity futures returns.

This table reports the estimated results of the pooled regression models of commodity futures returns on positive and negative news sentiment and their first five lags. The results are displayed for three new groups: all news, novel news and old news. This study conducts the regression for each new group, including the control variables, commodity fixed effect, year fixed effect, and day-of-week fixed effect. The standard errors of coefficients are clustered by the time dimension.

VARIABLES	All news		Novel news		Old news	
	Positive	Negative	Positive	Negative	Positive	Negative
	ret (t)	ret (t)	ret (t)	ret (t)	ret (t)	ret (t)
News sentiment (t)	0.245*** (11.924)	−0.237*** (−10.937)	0.092*** (8.510)	−0.084*** (−7.221)	0.234*** (11.203)	−0.261*** (−13.363)
News sentiment (t−1)	−0.020 (−1.520)	0.039** (2.642)	0.002 (0.160)	−0.001 (−0.121)	−0.021* (−1.825)	0.047*** (3.452)
News sentiment (t−2)	−0.041*** (−3.890)	0.014 (1.086)	−0.008 (−0.962)	−0.003 (−0.381)	−0.042*** (−3.747)	0.015 (1.236)
News sentiment (t−3)	−0.021** (−2.233)	0.006 (0.619)	0.000 (0.029)	0.002 (0.302)	−0.007 (−0.665)	0.019 (1.315)
News sentiment (t−4)	−0.032*** (−3.915)	−0.001 (−0.137)	0.006 (0.915)	−0.002 (−0.232)	−0.025*** (−2.939)	0.010 (1.033)
News sentiment (t−5)	−0.016 (−1.225)	0.024* (2.069)	−0.008 (−0.948)	−0.012 (−1.657)	−0.023* (−1.830)	0.023** (2.250)
Return (t−1)		−0.067*** (−10.715)		−0.026*** (−3.641)		−0.070*** (−10.774)
R-squared	0.097		0.060		0.099	

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5% and 10% levels, respectively.

Table 12

The effect of positive and negative news sentiment on commodity returns in the following 30 days.

This table reports the aggregated impacts of positive and negative news sentiment on returns of commodity futures for each 5-day period from the day $t-1$ to the day $t-30$, and for the period from the day t to the day $t-30$. This study conducts the regression with contemporaneous positive and negative news sentiment and their first 30 lags. The results are displayed for three new groups: all news, novel news, and old news. The total sentiment impact, the corresponding t -statistics, and significance are calculated from the Wald test. Each model is displayed in two columns, with the first for positive news sentiment and the second for negative news sentiment.

	All news		Novel news		Old news	
	Positive	Negative	Positive	Negative	Positive	Negative
News sentiment (t)	0.253*** (10.721)	−0.244*** (−10.718)	0.094*** (8.391)	−0.086*** (−7.350)	0.240*** (10.717)	−0.268*** (−12.794)
Total sentiment effect (t−5 to t−1)	−0.109*** (−4.461)	0.049** (2.641)	−0.004 (−0.188)	−0.020 (−1.354)	−0.090*** (−4.497)	0.089*** (7.989)
Total sentiment effect (t−10 to t−6)	−0.034 (−1.162)	0.028 (1.380)	−0.005 (−0.272)	−0.002 (−0.120)	−0.039 (−1.538)	0.018 (0.930)
Total sentiment effect (t−15 to t−11)	−0.016 (−0.511)	0.014 (0.415)	0.010 (0.597)	0.023 (1.450)	−0.021 (−0.654)	0.006 (0.181)
Total sentiment effect (t−20 to t−16)	−0.038* (−1.800)	0.009 (0.325)	−0.035** (−2.129)	−0.003 (−0.212)	−0.054** (−2.300)	−0.005 (−0.227)
Total sentiment effect (t−25 to t−21)	0.015 (0.396)	0.065* (2.013)	−0.006 (−0.474)	0.016 (1.416)	0.039 (1.264)	0.089*** (3.366)
Total sentiment effect (t−30 to t−26)	−0.013 (−0.440)	0.005 (0.189)	0.004 (0.248)	0.004 (0.295)	−0.018 (−0.822)	0.002 (0.081)
Total sentiment effect (t−30 to t)	0.058*** (3.327)	−0.075*** (−4.326)	0.058** (2.489)	−0.068*** (−3.146)	0.057*** (3.317)	−0.069*** (−4.093)

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5% and 10% levels, respectively.

positive and negative news sentiment and summarize the aggregated effects for each 5-day period and the total effect from day $t-30$ to day t in Table 12, similar to Table 9.

Table 12 reveals that only positive novel news sentiment shows a significant reversal of contemporaneous correlation for the time from day $t-20$ to day $t-16$. Other aggregated positive novel news sentiment effects for other 5-day periods are markedly more minor than the contemporaneous correlation and for the aggregated effect for the period from day $t-20$ to day $t-16$. We observe no significant reversals for any 5-day reported periods for negative novel news sentiment. For the old news sentiment, we observe strong and statistically significant reversals for both positive and negative news sentiment in the first five days. These adjustments are similar in magnitude for these two sides of news sentiment. Other significant reversals happen from day $t-20$ to day $t-16$ for positive old news sentiment and from day $t-25$ to day $t-21$. However, except for the time window of the first five days and

from day $t-25$ to day $t-21$, all other five-day time windows present much stronger reversals in magnitude for positive news than negative news sentiment. Although the reversals are mostly insignificant at these time windows, the t -statistics of coefficients of lagged positive old news sentiment are still markedly higher than that of lagged negative old news sentiment. This supports the above implication that the price-correcting process following overreaction to positive news sentiment spreads more than negative news sentiment.

Table 12 also reveals that the aggregated effect of positive novel news sentiment is similar to that of positive old news sentiment, at 0.06%. The aggregated effect of negative novel news sentiment is also at the same magnitude as that of negative old news sentiment, at 0.07%. These results suggest that investors' overreaction to the sentiment of old information in news items is mostly reversed within 30 days. Hence, we observe that the aggregated effects of positive and negative news sentiment move to the same value for both novel and

Table 13

Contemporaneous impact of news sentiment on the first and second nearby contracts. This table reports the impact of contemporaneous news sentiment on the first and second nearby contracts. First is a dummy variable that takes the values 1 and 0 for the first and second nearby contracts, respectively. To quantify the difference in the impact of news sentiment on the first and second nearby contracts, we interact the novel and old news sentiment variables with the First dummy. The analysis is conducted in three models: Model 1 focuses only on novel news sentiment, Model 2 exclusively on old news sentiment, and Model 3 incorporates both novel and old news sentiment. The standard errors of coefficients are clustered using the time dimension.

Model	1	2	3
VARIABLES	Return (t)	Return (t)	Return (t)
Novel news sentiment (t)	0.131*** (12.449)		0.076*** (8.978)
Novel news sentiment (t) x First	0.009*** (7.469)		0.006*** (4.928)
Old news sentiment (t)		0.353*** (15.046)	0.340*** (14.867)
Old news sentiment (t) x First		0.018*** (8.356)	0.017*** (7.809)
Control variables	Yes	Yes	Yes
Commodity and Time fixed effects	Yes	Yes	Yes
Observations	115,413	115,413	115,413
R-squared	0.061	0.094	0.095

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5% and 10% levels, respectively.

old news after 30 days, implying that the sentiment of old information triggers no further significant impact.

4.4. News sentiment impact on first and second nearby contracts

In this section, we test whether the impact of novel and old news sentiment varies on first and second nearby contracts. We extend our sample to include both the first and second nearby contracts and generate a first nearby dummy variable, which equals 1 and 0 for the first and second nearby contracts, respectively. We also interact our news sentiment variables with the first nearby dummy. The results are reported in Table 13.

The results indicate that both interactive terms are statistically significant, suggesting that the impact of old and novel news sentiment are stronger for the first nearby contract than for the second. This is plausible, as the first nearby contract likely receives more attention and is thus more affected by news sentiment. Specifically, the impact of novel news sentiment on the first nearby contract is 0.006 higher than for the second, while the impact of old news sentiment is 0.017 higher. These findings suggest that the impact of news sentiment on the first nearby contract is more pronounced for old news than for novel news sentiment.

We further extend this analysis to the lag of news sentiment. If the impact of news sentiment, especially old news sentiment, is more behavioral, we expect that the overreaction and reversal would be more pronounced for the first nearby contract. To investigate this, we include the first five lags of news sentiment in the model of returns. To examine the difference in the impact of news sentiment on returns between the first and second nearby contracts, we also add interactive terms between news sentiment and the first nearby dummy. We conduct three regression models. Model 1 considers only novel news:

$$r_{i,t} = \alpha + \sum_{j=0}^k \beta_k^{novel} Sent_{i,t-j}^{novel} + \sum_{j=0}^k \theta_k^{novel} Sent_{i,t-j}^{novel} \cdot First + \lambda First + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (10)$$

Model 2 considers only old news:

$$r_{i,t} = \alpha + \sum_{j=0}^k \beta_k^{old} Sent_{i,t-j}^{old} + \sum_{j=0}^k \theta_k^{old} Sent_{i,t-j}^{old} \cdot First + \lambda First + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (11)$$

And Model 3 considers both novel news and old news:

$$r_{i,t} = \alpha + \sum_{j=0}^k \beta_k^{novel} Sent_{i,t-j}^{novel} + \sum_{j=0}^k \theta_k^{novel} Sent_{i,t-j}^{novel} \cdot First + \sum_{j=0}^k \beta_k^{old} Sent_{i,t-j}^{old} + \sum_{j=0}^k \theta_k^{old} Sent_{i,t-j}^{old} \cdot First + \lambda First + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (12)$$

Table 14 reports the results of these models. The results in Table 14 align with those in Table 13, showing that the impact of both novel and old news sentiment are stronger for the first nearby contract. For novel news, both Model 1 and Model 3 indicate that only the impact of the third lag of novel news sentiment is significantly stronger for the first nearby contract, while other lags among the first five are not different between the first and second nearby contracts. In contrast, the impact of old news sentiment is significantly stronger for the first and second lags. We also observe a significantly stronger impact of the fourth lag of old news sentiment. Our analysis also reveals that the first nearby contracts, which receive more attention, have stronger overreactions to old news sentiment and subsequent reversals compared to the second nearby contracts.

4.5. News sentiment effect under financial distress and uncertainty

The literature has provided evidence that asset returns are more sensitive to sentiment during periods of high financial stress and uncertainty. Birru and Young (2022) investigate the relationship between sentiment and uncertainty and find that the impact of sentiment is notably stronger during times of higher uncertainty. This pattern is observed at both daily and yearly levels. García (2013) examine the impact of news sentiment during recession periods and also find a significantly higher effect of news sentiment in these stressful times. To test whether the relationship between commodity futures returns and news sentiment is amplified during periods of higher financial stress and uncertainty, we perform regressions of commodity futures returns on novel news sentiment, old news sentiment, moderators reflecting financial distress or uncertainty, and the interactive terms between these moderators and novel and old news sentiment. As the previous results indicate that old news sentiment correlates with asset returns much more strongly than novel news sentiment, we expect returns to be more correlated with old news sentiment than novel news sentiment as levels of financial distress and uncertainty increase.

We use the VIX index to reflect financial uncertainty in the market and employ the financial stress indices from the US Government Office of Financial Research (OFR FSI) and the Federal Reserve Bank of St. Louis (STL FSI). The Office of Financial Research also provides five categorical stress indices. The Credit Index measures the variance in borrowing costs across firms, highlighting shifts in credit market conditions. The Equity Valuation Index reflects investor confidence and risk tendencies through diverse stock market valuations. We also use the Funding Index to understand the funding accessibility for financial institutions, especially under market stress. The Safe Assets Index tracks the value of safe assets, revealing investor shifts during volatile times. Lastly, the Volatility Index provides a comprehensive view of market stability, capturing volatility across various asset classes, including equity, credit, currency, and commodity markets. We also examine the impact of regional financial stress via the financial stress indices for the US, advanced economies, and emerging economies provided by the Office of Financial Research. We conduct the tests using the following regression

$$r_{i,t} = \alpha + \beta_1 Sent_{i,t}^{novel} + \beta_2 Sent_{i,t}^{old} + \theta_1 Sent_{i,t}^{novel} \times Mod + \theta_2 Sent_{i,t}^{old} \times Mod + \eta Mod + \delta r_{i,t-1} + \sum \gamma_q Contr_{q,t} + F_{i,t} + u_{i,t} \quad (13)$$

In the regression, *Mod* refers to each index among the uncertainty and financial stress indices. $Sent_{i,t}^{novel}$ and $Sent_{i,t}^{old}$ are novel and old

Table 14

The lagged impact of news sentiment on first and second nearby contracts.

This table examines the differential impact of news sentiment on first and second nearby contracts. First is a dummy variable that takes the values 1 and 0 for the first and second nearby contracts, respectively. The model incorporates the first five lags of news sentiment and interaction terms between sentiment variables and the First dummy. We run three models: Model 1 for novel news, Model 2 for old news, and Model 3 for both novel and old news. For Models 1 and 2, results are presented in two columns: the first shows estimated coefficients of news sentiment, while the second displays coefficients of the interaction term between sentiment variables and the first nearby contract dummy. Model 3's results span four columns: two for novel news sentiment coefficients and two for old news sentiment coefficients.

Model	1		2		3			
	Novel news		Old news		Novel news		Old news	
VARIABLES	Sent (t)	Sent (t) × First	Sent (t)	Sent (t) × First	Sent (t)	Sent (t) × First	Sent (t)	Sent (t) × First
sentiment (t)	0.131*** (12.065)	0.009*** (7.695)	0.391*** (15.332)	0.022*** (9.074)	0.079*** (9.199)	0.006*** (5.269)	0.378*** (15.062)	0.021*** (8.505)
sentiment (t-1)	0.004 (0.398)	-0.001 (-1.349)	-0.051*** (-6.154)	-0.006*** (-4.205)	-0.005 (-0.509)	-0.002 (-1.649)	-0.051*** (-6.089)	-0.005*** (-3.947)
sentiment (t-2)	-0.001 (-0.150)	-0.002 (-1.495)	-0.046*** (-5.197)	-0.004*** (-3.884)	-0.004 (-0.481)	-0.002 (-1.415)	-0.045*** (-5.215)	-0.003*** (-3.630)
sentiment (t-3)	0.000 (0.066)	-0.002** (-2.300)	-0.022** (-2.460)	-0.002 (-1.234)	-0.004 (-0.530)	-0.002** (-2.304)	-0.022** (-2.480)	-0.001 (-1.092)
sentiment (t-4)	0.005 (0.907)	0.002 (1.173)	-0.027** (-2.826)	-0.003*** (-3.256)	0.004 (0.862)	0.002 (1.470)	-0.027** (-2.731)	-0.004*** (-3.343)
sentiment (t-5)	0.003 (0.436)	0.000 (0.394)	-0.037*** (-3.683)	-0.001 (-1.263)	-0.001 (-0.163)	0.000 (0.307)	-0.040*** (-4.037)	-0.001 (-1.582)
Observations	115,283		115,283		115,283			
R-squared	0.061		0.098		0.100			

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5% and 10% levels, respectively.

news sentiment of commodity i on the day t . The moderating effect of uncertainty and financial stress is reflected through the coefficients of interactive terms between each index and the sentiment of novel and old news. We run a separate regression for each index and regional index and report the results in Table 15. Panel A reports the findings using contemporaneous moderators, whereas Panel B includes the results using the first lag of the moderators. Consistent with previous tables, the results indicate a positive correlation between novel and old news sentiment and returns of commodity futures. Additionally, the findings also confirm that commodity futures returns are more responsive to changes in old news sentiment compared to novel news sentiment.

Panel A demonstrates that the interaction between each moderator and old news sentiment has a higher coefficient compared to the interaction between the same moderator and novel news sentiment. Specifically, when there is higher stock market uncertainty (proxied by VIX), commodity futures returns are more sensitive to old news sentiment rather than novel news sentiment (0.006 compared to 0.003). The results using financial stress indicators also highlight the moderating effect of financial stress on the relationship between old news sentiment and returns. Although the coefficient of the interactive term between novel news sentiment and stress indices is statistically significant at 0.007 (t-stat = 2.349) and 0.027 (t-stat = 2.953) when using general measures of financial stress, OFR FSI and STL FSI, respectively, they are only half of the corresponding interaction terms between old news sentiment and these financial stress indices (0.013 with t-stat = 5.209 for OFR FSI and 0.042 with t-stat = 5.299 for STL FSI). This suggests that under higher financial stress, commodity futures returns are more correlated with old news sentiment than with novel news sentiment, indicating that the influence of old news sentiment is intensified during periods of increased uncertainty and financial stress.

Regarding regional stress, the results show that commodity futures returns are more responsive to old news sentiment than novel news sentiment when financial stress is higher in the US or in advanced economies. The coefficient of old news sentiment is similar in Models 8 and 9 (around 0.359), but the coefficient of the interactive term between old news sentiment and US financial distress (0.035) is higher than that between old news sentiment and advanced economies' financial distress (0.026). However, the interactive term between old news sentiment and emerging market stress is not significant. These findings emphasize the influence of advanced economies, particularly the US

market, on the response of the commodity futures market to media news sentiment.

We also report the results for each category of stress from the Office of Financial Research in Table 16. In each category of financial stress, all the interactive terms between old news sentiment and credit, equity, safe assets, funding, and volatility stress indices (0.030, 0.100, 0.118, 0.051, and 0.040, respectively) are statistically significant at the 1% level. However, only the interactive term between novel news sentiment and funding stress (0.030) is significant at the 1% level, indicating that access to funding of financial institutions is an important factor that impacts investor responses to sentiment in media news. The interactive terms between novel news sentiment and credit, equity, and volatility (0.019, 0.053, 0.017, respectively) are only significant at the 10% level, whereas there is no moderating effect of safe asset stress on the impact of novel news sentiment. Among these five stress indices, credit and funding stress strongly influence the relationship between novel news sentiment and commodity futures returns. Although credit and funding stress also affect the relationship between old news sentiment and returns, their effects are less pronounced compared to the impact of safe assets on old news sentiment.

Panel B of Tables 15 and 16 present the results when using lagged moderators instead of contemporaneous moderators. The findings in these tables are consistent with those in Panel A. The moderating effects of financial uncertainty and stress are more pronounced for old news sentiment, with all the interactive terms with old news sentiment being statistically significant at the 1% level, except for the case of emerging market stress. Panel B also indicates that the interactive terms between novel news sentiment and lagged moderators are less significant than those between novel news sentiment and contemporaneous moderators in Panel A, except for the case of funding stress. This suggests that the impact of financial stress and uncertainty on the relationship between novel news sentiment and returns weakens quickly over time, whereas the impact on the relationship between old news sentiment and returns is more persistent.

5. Conclusion

In this research, we investigate the relationship between news sentiment and commodity futures returns. Our findings reveal several key insights. We find a statistically significant correlation between news

Table 15

News sentiment effect under financial distress and uncertainty.

This table presents the moderating effect of uncertainty (measured by VIX) and financial stress (measured by the Financial Stress Index) from the US Government Office of Financial Research (OFR FSI) and the Federal Reserve Bank of St. Louis (STL FSI). The financial stress indices from the Office of Financial Research include three regional stress indices that reflect financial stress in the US (US), Advanced economies (Advanced), and Emerging economies (Emerging). To examine the moderating effect, this study conducts regressions of commodity futures returns on news sentiment, the moderator, and the interactive terms between news sentiment and the moderator. Panel A shows the results using contemporaneous moderators, whereas Panel B shows the results using the first lag of the moderators. The standard errors of coefficients are clustered by the time dimension.

Panel A: Models using contemporaneous moderators						
Model	1	2	3	4	5	6
Dependent variable	Ret (t)	Ret (t)	Ret (t)	Ret (t)	Ret (t)	Ret (t)
Variables\Moderators	VIX	OFR FSI	US	Advanced	Emerging	STL FSI
Novel news sentiment (t)	0.028 (0.969)	0.085*** (10.705)	0.084*** (10.583)	0.085*** (10.494)	0.086*** (9.645)	0.083*** (11.478)
Novel news sentiment (t) * Moderator (t)	0.003 (1.709)	0.007** (2.349)	0.017** (2.399)	0.014** (2.205)	0.052 (1.682)	0.027*** (2.953)
Old news sentiment (t)	0.243*** (6.260)	0.360*** (17.339)	0.359*** (18.394)	0.359*** (16.819)	0.361*** (15.321)	0.355*** (16.052)
Old news sentiment (t) * Moderator (t)	0.006*** (4.633)	0.013*** (5.209)	0.035*** (6.193)	0.026*** (4.741)	0.075 (1.673)	0.042*** (5.299)
Observations	57,729	57,729	57,729	57,729	57,729	57,729
R-squared	0.095	0.096	0.096	0.096	0.095	0.096
Panel B: Models using the first lag of moderators						
Model	1	2	3	4	5	6
Dependent variable	Ret (t)	Ret (t)	Ret (t)	Ret (t)	Ret (t)	Ret (t)
Variables\Moderators	VIX	OFR FSI	US	Advanced	Emerging	STL FSI
Novel news sentiment (t)	0.040 (1.117)	0.085*** (10.534)	0.084*** (10.468)	0.085*** (10.386)	0.086*** (9.423)	0.083*** (11.350)
Novel news sentiment (t) * Moderator (t-1)	0.002 (1.084)	0.007* (2.045)	0.016* (2.014)	0.013* (1.987)	0.049 (1.532)	0.024** (2.537)
Old news sentiment (t)	0.257*** (6.661)	0.359*** (17.342)	0.359*** (18.358)	0.359*** (16.866)	0.361*** (15.277)	0.354*** (16.020)
Old news sentiment (t) * Moderator (t-1)	0.005*** (4.696)	0.012*** (4.862)	0.033*** (5.978)	0.023*** (4.369)	0.066 (1.513)	0.041*** (5.226)
Observations	57,729	57,729	57,729	57,729	57,729	57,729
R-squared	0.095	0.096	0.096	0.096	0.095	0.096

Coefficients with “***”, “**”, and “*” indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

sentiment and commodity futures returns. Additionally, both novel and old news sentiment are associated with higher returns, with old news sentiment having a stronger impact. These findings support the theory of investor inattention (Gilbert et al., 2012), which suggests that investors are often confused between old and new information. As a result, it is difficult for investors to differentiate between the sentiment of old news and novel news. We also explore the effect of positive and negative news sentiment. We observe that positive news sentiment is linked to an increase in returns, whereas negative news sentiment is linked to a decrease in returns. These results align with the impact pattern of news sentiment in García (2013) for the equity market and in Smales (2014) for gold commodity futures.

Furthermore, we find that the sentiment of novel news does not trigger significant overreaction in the commodity futures market. However, the sentiment of old news triggers overreaction on the news day, followed by return reversals. This aligns with the investor inattention theory, highlighting the tendency of investors to overreact to old information combined with novel news. Additionally, the effects of both novel and old news sentiment become similar after 30 trading days, suggesting that overreaction to old news sentiment is mostly reversed within this period. Our results show a more pronounced overreaction of commodity futures returns to old news sentiment and subsequent reversals for the first nearby contracts compared to the second nearby contracts.

Lastly, this study examines the impact of news sentiment on commodity futures returns during periods of high financial stress and uncertainty. Using various stress and uncertainty indices, including the VIX index and financial stress indices from the US Government Office of Financial Research and the Federal Reserve Bank of St. Louis, we find

that old news sentiment has a stronger impact on returns compared to novel news sentiment in the time of higher financial stress and uncertainty. The impact of old news sentiment is more significantly intensified than that of novel news sentiment when financial stress components, such as credit stress, equity stress, safe assets stress, funding stress, and volatility, are elevated. Additionally, the findings indicate that commodity futures returns are more responsive to old news sentiment when financial stress is higher in the US or in advanced economies, but there is no significant moderating impact of stress from emerging markets.

Overall, our research provides insights into the importance of news sentiment, particularly old news sentiment, in driving the response of commodity futures returns. These insights provide helpful implications for market participants in understanding the news sentiment signals when investigating the asset price in the commodity futures market.

CRediT authorship contribution statement

Yeguang Chi: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lina El-Jahel:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Thanh Vu:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Table 16

News sentiment effect under financial distress and uncertainty.

This table presents the moderating effect of financial stress components from the US Government Office of Financial Research (OFR FSI). The financial stress indices from the Office of Financial Research include five categorical indices: credit spreads (credit), equity valuation (equity), funding access of financial institutions (funding), value of safe assets (safe assets), and volatility of equity, credit, currency, and commodity markets (volatility). To examine the moderating effect, this study conducts regressions of commodity futures returns on news sentiment, the moderator, and the interactive terms between news sentiment and the moderator. Panel A shows the results using contemporaneous moderators, whereas Panel B shows the results using the first lag of the moderators. The standard errors of coefficients are clustered by the time dimension.

Panel A: Models using contemporaneous moderators					
Model	1	2	3	4	5
Dependent variable	Ret (t)	Ret (t)	Ret (t)	Ret (t)	Ret (t)
Variables\Moderators	Credit	Equity	Safe assets	Funding	Volatility
Novel news sentiment (t)	0.082*** (11.524)	0.087*** (10.395)	0.081*** (10.691)	0.079*** (11.445)	0.091*** (7.662)
Novel news sentiment (t) * Moderator (t)	0.019* (1.979)	0.053* (1.839)	0.000 (0.010)	0.030*** (4.203)	0.017* (1.741)
Old news sentiment (t)	0.354*** (15.268)	0.366*** (16.427)	0.349*** (15.947)	0.350*** (16.959)	0.375*** (18.899)
Old news sentiment (t) * Moderator (t)	0.030** (2.763)	0.100*** (5.232)	0.118*** (3.591)	0.051*** (5.706)	0.040*** (5.132)
Observations	57,729	57,729	57,729	57,729	57,729
R-squared	0.095	0.095	0.095	0.096	0.096
Panel B: Models using the first lag of moderators					
Model	1	2	3	4	5
Dependent variable	Ret (t)	Ret (t)	Ret (t)	Ret (t)	Ret (t)
Variables\Moderators	Credit	Equity	Safe assets	Funding	Volatility
Novel news sentiment (t)	0.082*** (11.427)	0.087*** (9.745)	0.082*** (10.696)	0.079*** (11.412)	0.090*** (7.353)
Novel news sentiment (t) * Moderator (t-1)	0.018* (1.896)	0.046 (1.457)	-0.008 (-0.214)	0.031*** (4.232)	0.015 (1.432)
Old news sentiment (t)	0.354*** (15.292)	0.365*** (16.269)	0.349*** (15.947)	0.350*** (16.956)	0.372*** (18.884)
Old news sentiment (t) * Moderator (t-1)	0.029** (2.709)	0.089*** (4.827)	0.113*** (3.602)	0.049*** (5.544)	0.035*** (4.553)
Observations	57,729	57,729	57,729	57,729	57,729
R-squared	0.095	0.095	0.095	0.096	0.096

Coefficients with “***”, “**”, and “*” indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A.1

Key fields in TRNA flat news fields data.

This table provides the main data fields in the file “Flat news fields”, one of two data types provided in TRNA data. This file records all news related to commodities. Each line of data corresponds to one news and when it was fed into the system. The fields in this file summarize the features and background information of the news.

Field	Meaning
ID	The id coded for the news in the system
Headline	The title of the news
FeedTimestamp	The time-stamp that the news is fed in the system. The lag between the fed time and the actual released time of the news is not greater than 500 ms. Time is quoted in UTC time zone
Language	The language of the news
BodySize	The number of characters in the news
CompanyCount	The number of commodities mentioned in the news
SentenceCount	The number of sentences in the news
WordCount	The number of tokens in the news. Tokens include both words and punctuation

Appendix

Key fields in Thomson Reuters News Analytics data

This study employed unique data from Thomson Reuters: Thomson Reuters News Analytics (TRNA) featured by Thomson Reuters Machine

Readable News. This platform covers news from Reuters and other regulatory news services to feed timely news of about 35,000 equities and 40 commodities. TRNA provides some key features of the news for the commodity market, as shown in Table A.2. In this study’s scope, we focus primarily on sentiment data and the relevance of news to the commodities. TRNA covers 43 active IDs related to commodities.

TRNA data for commodity markets consists of two types of reported files. One pertains to news-specific information, referred to as “Flat news fields”, whereas the other, called “Flat scores field”, provides asset-specific analysis. The “Flat news fields” file contains descriptions for the news that is fed into the system. Each row in this news-specific data corresponds to one news item in the system. The main fields reported in this file are explained in Table A.1.

The second type of files in the TRNA data provides news analysis, which is of particular interest in this study due to its inclusion of sentiment analysis and its potential for calculating media news sentiment on specific commodities. Each row in this file corresponds to a single commodity and a new item mentioned in the news feed system. The “Flat scores fields” contain significant information, as documented in Table A.2.

Appendix table

See Tables A.1–A.5

Table A.2

Key fields in TRNA flat score fields data.

This table provides the main data fields in the file “Flat score fields”, one of two data types provided in TRNA data. This file contains sentiment calculations from textual analysis of the news listed in the file “Flat news field”. Each row in this file is the values of fields computed for the commodity mentioned in a piece of news. Therefore, a single line in the “Flat news fields” file will correspond to multiple lines in the “Flat score news” file because a single piece of news might mention different commodities. In addition to the sentiment calculations, this file also provides information about the relevance of the news to the commodity under consideration, the novelty of the news and other information about the news used to link to the file “Flat news field” as needed.

Field	Meaning
FeedTimestamp	The timestamp when the news is fed in the system. This field can be used to merge the score file with the news file.
AssetCodes/Asset Id	The code of the commodities reported in the news.
AssetName	The name of the commodities corresponding to the AssetCodes/Asset Id.
FirstMentionSentence	The first sentence that the reported commodity is mentioned. The data starts to count from the title as the first sentence. This information can be used as a proxy for the relevant level of the commodity to the news in the aspect that more relevant commodities are mentioned earlier in the text. If this field takes the value of 0, it means that the reported commodities are not mentioned directly in the news. Therefore, all content of the news will be used to calculate sentiment scores.
Relevance	The relevant level of the mentioned commodity to the news. It takes the value from 0 to 1, which is determined by comparing the number of occurrences of the commodity to those of other assets also mentioned in the news. If the name of the commodity appears in the headline of the news, the relevance is set to 1.
SentimentClass	The dominating tone of news. It takes one of three values: -1, 0, 1 corresponding to whether the news is more negative, neutral or positive.
SentimentNegative	The probability that the news presents a negative tone to the commodity.
SentimentPositive	The probability that the news presents a positive tone to the commodity.
SentimentNeutral	The probability that the news presents a neutral tone to the commodity, which is neither positive or negative. The data considers only three aspects of news tone, so the sum of all the above sentiment probability is equal to 1.
NoveltyCount	This is an array containing five numbers that reflect the similarity of the news to the prior news in five-time windows: 12 h, 24 h, three days, five days and a calendar week. For each time window, the news will be compared with other news to check whether it is similar in linguistic aspect to other news released in that time window prior to the reported news. The value for each time window is the number of similar news. If this field takes the value of 0 for a time window, it means that the content of news related to the mentioned commodity is considered novel to all the news released within the time window before the news releases

Table A.3

Specifications of the selected commodity futures.

This table provides the specifications of the futures contracts selected in the sample. CRB ticker is the ID number of the contracts on the respective exchanges. The name of the contract is the official name of the futures contracts on the exchanges, and the contract size is the volume of the corresponding commodity in each contract unit.

Commodity name	Type	CRB ticker	Name of contract	Contract size	Exchange
Crude Oil	Energy	CL	Crude Oil West Texas Intermediate	1,000 U.S. barrels (42,000 gallons)	NYMEX
Gasoline	Energy	RB	Gasoline Blendstock New York Harbor	42,000 gallons	NYMEX
Natural Gas	Energy	NG	Natural Gas	10,000 MMBtu	NYMEX
Corn	Grains	C	Corn (pit)	5,000 bushels	CBOT
Soybeans	Grains	S	Soybean (pit)	5,000 bushels	CBOT
Wheat	Grains	W	Chicago Soft Red Winter Wheat	5,000 bushels	CBOT
Copper	Metals	HG	High Grade Copper	25,000 pounds	COMEX
Gold	Metals	GC	Gold 100-oz	100 fine troy ounces	COMEX
Cocoa	Softs	CC	Cocoa	10 metric tonnes	ICE
Coffee	Softs	KC	Coffee C Arabica	37,500 pounds (~250 bags)	ICE
Cotton	Softs	CT	Cotton #2	50,000 pounds (~100 bales)	ICE
Orange Juice	Softs	OJ	Orange Juice [FCOJ-A]	15,000 pounds	ICE
Sugar	Softs	SB	Sugar #11	112,000 pounds (50 long tonnes)	ICE

Table A.4

Robustness test - Different clustering settings.

This table presents a robustness test for the results in Table 6 by changing the clustering dimension for standard errors. In Table 6, we correct standard errors using the time dimension due to the weak autocorrelation of commodity futures returns. In this table, we report the results when clustering by commodity dimension and by both commodity and time dimensions. Models 1 and 2 show the results for news sentiment of all news, while Models 3 and 4 present the results for news sentiment of novel and old news.

	(1)	(2)	(3)	(4)
VARIABLES	Return (t)	Return (t)	Return (t)	Return (t)
All news sentiment (t)	0.370*** (7.807)	0.370*** (7.578)		
Novel news sentiment (t)			0.082*** (7.987)	0.082*** (7.244)
Old news sentiment (t)			0.358*** (7.580)	0.358*** (7.341)

(continued on next page)

Table A.4 (continued).

	(1)	(2)	(3)	(4)
VARIABLES	Return (t)	Return (t)	Return (t)	Return (t)
Return (t-1)	-0.066*** (-3.788)	-0.066*** (-3.911)	-0.069*** (-3.892)	-0.069*** (-4.003)
CRSP return (t)	0.204*** (4.421)	0.204*** (4.315)	0.204*** (4.413)	0.204*** (4.314)
Risk-free rate (t)	-21.223*** (-3.460)	-21.223*** (-3.166)	-20.725*** (-3.302)	-20.725** (-3.016)
VIX (t)	-0.003 (-1.233)	-0.003 (-1.383)	-0.003 (-1.207)	-0.003 (-1.346)
TED spread (t)	-0.084** (-2.402)	-0.084 (-0.825)	-0.081** (-2.361)	-0.081 (-0.776)
D Usdollar (t)	-0.631*** (-11.412)	-0.631*** (-5.673)	-0.628*** (-11.324)	-0.628*** (-5.620)
Constant	0.181** (2.745)	0.181** (2.710)	0.177** (2.617)	0.177** (2.598)
SE clustered by commodity	Yes	Yes	Yes	Yes
SE clustered by time	No	Yes	No	Yes
Observations	57,729	57,729	57,729	57,729
R-squared	0.092	0.092	0.094	0.094

Coefficients marked with “***”, “**”, and “*” are statistically significant at the 1%, 5%, and 10% levels, respectively.

Table A.5

Robustness test - Adding fundamental factors.

This table reports a robustness test for the results in Table 6 by incorporating additional control variables: basis, momentum, basis-momentum, volatility, and liquidity. We calculate basis, momentum, and basis-momentum using the formulas from Boons and Prado (2019), adapted for daily frequency. Volatility is measured as the standard deviation of returns over the past 30 days. The liquidity factor is calculated as the natural logarithm of trading volume plus one, standardized for each commodity. Models 1 and 3 report the test using only key control variables in Table 6, while Models 2 and 4 add basis, momentum, basis-momentum, volatility, and liquidity to Models 1 and 3, respectively. We also report the test for the difference between the impact of novel and old news in the last two rows of the table. The coefficients are clustered by time dimension.

	(1)	(2)	(3)	(4)
VARIABLES	Return (t)	Return (t)	Return (t)	Return (t)
All news sentiment (t)	0.370*** (15.244)	0.375*** (14.988)		
Novel news sentiment (t)			0.082*** (8.970)	0.088*** (9.519)
Old news sentiment (t)			0.358*** (14.600)	0.361*** (14.450)
Return (t-1)	-0.066*** (-10.902)	-0.062*** (-9.084)	-0.069*** (-11.092)	-0.065*** (-9.361)
CRSP return (t)	0.204*** (8.478)	0.211*** (8.312)	0.204*** (8.522)	0.211*** (8.374)
Risk-free rate (t)	-21.223** (-2.489)	-25.948** (-2.376)	-20.725** (-2.418)	-25.464** (-2.331)
VIX (t)	-0.003* (-1.790)	-0.006*** (-3.184)	-0.003* (-1.764)	-0.006*** (-3.136)
TED spread (t)	-0.084 (-0.779)	-0.097 (-0.813)	-0.081 (-0.731)	-0.093 (-0.766)
D Usdollar (t)	-0.631*** (-6.035)	-0.622*** (-5.801)	-0.628*** (-5.979)	-0.619*** (-5.749)
Basis (t-1)		0.642 (1.260)		0.699 (1.346)
Momentum (t-1)		-0.338 (-1.284)		-0.351 (-1.312)
Basis-momentum (t-1)		4.040*** (6.526)		3.898*** (6.173)
Volatility (t-1)		-0.133* (-1.943)		-0.127* (-1.837)
Liquidity (t-1)		0.009 (0.738)		0.009 (0.715)
Constant	0.148** (2.288)	0.178** (2.345)	0.151** (2.248)	0.179** (2.279)
Observations	57,729	55,280	57,729	55,280
R-squared	0.092	0.092	0.094	0.094
Old - Novel news sentiment t-stat			0.276 11.067	0.273 11.054

Coefficients with “***”, “**”, and “*” are statistically significant at 1%, 5%, and 10% levels, respectively.

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